

ORIGINAL RESEARCH

Prevalence and anatomical distribution of MRI lesions in axSpA and differences between patients with and without peripheral involvement: results from the ASAS classification cohort

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ABSTRACT

Objective To evaluate the prevalence and anatomical distribution of inflammatory and structural MRI lesions in axial spondyloarthritis (axSpA) and compare these between patients with isolated axial involvement and those with peripheral manifestations.

Methods Data from the Assessment of SpondyloArthritis International Society (ASAS) Classification Cohort were analysed. Peripheral involvement was defined as past or current arthritis/dactylitis/enthesitis. Sacroiliac joint (SIJ) and spinal MRI lesions typical of axSpA were classified per ASAS lesion definitions and centrally read with multi-reader majority agreement (lesion present if called by the majority; SIJ $\geq 4/7$, spine $\geq 5/9$ readers). Comparisons between patients with and without peripheral manifestations were made.

Results Among 199 axSpA patients with SIJ MRI, 67 also had spinal MRI. Subchondral SIJ bone marrow oedema (BMO) was observed in 49%, without quadrant preference or subgroup differences. Other SIJ inflammatory lesions ranged from 4%–18%. Erosions (35%) and fat lesions (22%) were the most frequent structural lesions. In the spine, BMO, fat lesions and syndesmophytes/ankylosis were detected in 38%, 25% and 5%, respectively, with similar subgroup frequencies. Among 40 patients with both SIJ and whole spine MRI, inflammatory lesions were observed in both sites in 18%, SIJ only in 38%, and spine only in 20%. Structural lesions occurred in both sites in 19%, SIJ only in 30%, and spine only in 5%, with no subgroup differences.

Conclusion The prevalence and anatomical distribution of ASAS-defined MRI lesions was similar across axSpA subgroups. Notably, 20% exhibited spine-only inflammation, suggesting potential added diagnostic and monitoring value of spinal MRI, warranting further study.

INTRODUCTION

Axial spondyloarthritis (axSpA) is a chronic inflammatory disease primarily affecting the

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ MRI is crucial for detecting inflammatory and structural lesions in axial spondyloarthritis (axSpA), playing a central role in diagnosis and classification.
- ⇒ The prevalence and anatomical distribution of sacroiliac joint (SIJ) and spinal MRI lesions in axSpA, particularly in relation to peripheral involvement, remains poorly characterised.

WHAT THIS STUDY ADDS

- ⇒ This study provides a detailed mapping of inflammatory and structural MRI lesions in the SIJ and spine, applying ASAS definitions.
- ⇒ It demonstrates that axSpA patients, irrespective of peripheral involvement, exhibit similar lesion distributions and frequencies, with a notable subset showing spinal lesions in the absence of SIJ involvement.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ Findings may refine diagnostic approaches by enhancing understanding of MRI lesion patterns and their phenotypic overlap in axSpA.
- ⇒ A subset of patients demonstrated spinal lesions in the absence of sacroiliac joint involvement. These findings underscore the potential role of spinal MRI in complementing SIJ imaging assessment. Nevertheless, its diagnostic and monitoring value remains to be clearly defined, and robust prospective controlled studies are essential to establish its clinical utility.

axial skeleton, leading to active inflammation and structural damage in the sacroiliac joints (SIJs) and spine.^{1,2} Magnetic resonance imaging (MRI) has become an essential tool for axSpA assessment, enabling the

visualisation of both inflammatory lesions and structural changes. SIJ imaging is incorporated into the Assessment of SpondyloArthritis International Society (ASAS) classification criteria for axSpA as a key objective component, highlighting its relevance.³ In contrast, while spinal MRI is frequently performed in clinical practice, it has not yet been formally included in classification criteria. Nonetheless, spinal MRI may aid in diagnosing axSpA, particularly in cases where SIJ findings are inconclusive or absent, reinforcing its role in early disease identification and comprehensive assessment.^{3–10}

In the UK, the British Society for Spondyloarthritis consensus guidelines advocate MRI of the entire spine and SIJs, using T1-weighted (T1W) and fat-suppressed, fluid-sensitive sequences (including T2-weighted fat-suppressed (T2WFS), Short Tau Inversion Recovery or Dixon methods), to evaluate active and structural changes indicative of axSpA and assist in the diagnostic process.^{11,12} However, European Alliance of Associations for Rheumatology (EULAR) guidelines do not endorse routine MRI for monitoring treatment response or disease activity due to limited evidence and cost concerns.¹³ The ASAS and Spondyloarthritis Research and Treatment Network (SPARTAN) collaboration recently recommended incorporating a T1W thin slice sequence, such as Volumetric Interpolated Breath-hold Examination, for more reliable assessment of structural damage in the SIJs.¹⁴

To enhance the reliability of MRI assessment and diagnostic accuracy, ASAS has recently updated MRI lesion definitions for axSpA in both the SIJ⁴ and spine.⁵ Despite these refinements and MRI's central role in axSpA management, limited data exist on the anatomical distribution, frequency and lesion combinations in axSpA, particularly when stratified by clinical phenotypes. Moreover, the extent to which lesion patterns differ between clinical subgroups—specifically between patients with and without peripheral involvement—remains unclear. A deeper understanding of these variations could provide valuable insights into disease heterogeneity and inform diagnostic strategies.

This study aims to characterise the prevalence and anatomical distribution of inflammatory and structural MRI lesions in axSpA, based on ASAS definitions. Additionally, it explores differences between patients with peripheral involvement (axial-plus-peripheral subgroup) and those without (axial-only subgroup), using data from the ASAS Classification Cohort (ASAS-CC).³

METHODS

Study population and database

This study analysed data from the ASAS-CC.³ The ASAS-CC cohort was an international, multicentre, prospective study. Rheumatologists from 29 ASAS centres worldwide included 975 consecutive patients who first presented for diagnostic work-up. To be included, eligible patients had to fulfil one of two criteria: (1) 'axial population': chronic (>3 months) back pain of unknown origin (no definite

diagnosis) with an age of onset below 45 years, with or without peripheral symptoms and (2) 'peripheral population': patients with peripheral arthritis and/or enthesitis and/or dactylitis and the absence of current back pain with suspicion of spondyloarthritis (SpA) but no definitive diagnosis. Patients with a clinical diagnosis of axSpA at baseline who underwent SIJ and/or spinal MRI as part of the ASAS-CC study were included in our analysis. The cohort was divided into subgroups according to the absence or presence of peripheral involvement, which was defined as a history or present arthritis, dactylitis or enthesitis, yielding an axial-only group (none present) and an axial-plus-peripheral group (≥1 present).

Data collection

Baseline demographic, clinical and laboratory data were evaluated, including age, sex, human leucocyte antigen B27 (HLA-B27) status, history of uveitis, psoriasis, inflammatory bowel disease, elevated C-reactive protein (CRP), arthritis, dactylitis, enthesitis, family history of SpA and SIJ/spine MRI findings.

MRI lesions and definitions

MRI lesions were classified according to newly established ASAS definitions.^{4,5} The updated definitions, published in 2019 for the SIJs and in 2022 for the spine, were developed through a structured consensus process, followed by a validation study using MRI scans from patients enrolled in the ASAS-CC. As previously reported,^{4,5} the ASAS-MRI working group finalised the consensus wording for these definitions and curated a set of reference images illustrating each lesion type. In addition, the group designed the ASAS MRImagine study, which incorporated a multi-reader evaluation of lesion definitions and a study-specific interactive electronic case report form (eCRF). Data from the ASAS MRImagine study readings formed the basis for the present analysis.

This analysis was descriptive and anatomic in a cohort of patients with a diagnosis of axSpA. Scans were scored for ASAS-defined lesions. A key element of ASAS-defined lesions is that "the appearance of all lesions must be highly suggestive of spondyloarthritis. Although only scans from patients with a confirmed diagnosis of axSpA were included in the present analyses, readers also evaluated scans from non-axSpA cases in a blinded manner, as part of the ASAS MRImagine study. The eCRF permitted documentation of inflammation even when it was not typical of axSpA. If a scan was not suggestive of axSpA, one or more alternative diagnoses could be selected from predefined options: normal variation of the SIJs (SIJ), lumbosacral transitional anomaly, osteoarthritis/degeneration, osteitis condensans ilii, fracture/stress fracture, infection or previous infection, bone cyst, neoplasia (benign or malignant) or other (with specification).

Inflammatory SIJ lesions included subchondral bone marrow oedema (BMO), capsulitis, joint space enhancement (recorded only when postcontrast SIJ images were available and otherwise not assessed), inflammation at the

site of erosion, enthesitis and joint space fluid. Structural SIJ lesions encompassed erosions, fat lesions (fat metaplasia/deposition), fat metaplasia within erosion cavities (backfill), sclerosis, ankylosis and non-bridging bone buds.⁴ Spinal MRI lesions were categorised as inflammatory (BMO) and structural (fat lesions, erosions, bone spurs/syndesmophytes and ankylosis).⁵

Lesion locations were mapped: for the SIJ, lesions were reported in four side-specific regions—right ilium, left ilium, right sacrum and left sacrum—according to ASAS definitions. Spinal lesions were dichotomously recorded at each vertebral disc/vertebral unit level (C2-S1, distinguishing upper and lower localisations, and including all the following anatomical structures: vertebral bodies, facet joints, transverse processes, costovertebral and costotransverse joints, spinous processes and soft tissues/interspinous ligaments). For anatomical mapping, any qualifying lesion at a recorded location counted as present; region-level ‘SIJ involvement’ and ‘spine involvement’ indicate ≥ 1 qualifying lesion within the SIJ or spine, respectively. At each recorded location, lesions were classified as a binary variable (present/absent) in accordance with ASAS definitions, rather than as quantitative counts or scores.

Images were centrally evaluated by experienced ASAS MRI readers (seven for SIJ; nine for spine), each with over 10 years of reading experience. Prior to assessment, readers underwent calibration and were blinded to clinical data and subgroup allocation. Lesion presence was determined by majority agreement, requiring agreement from at least 4 out of 7 readers for SIJ lesions or 5 out of 9 readers for spinal lesions; no post hoc consensus adjudication was performed.

Statistical analyses

Inflammatory and structural lesions were evaluated separately for the entire axSpA cohort, with comparisons made between patients with peripheral involvement (axial-plus-peripheral subgroup) and those without (axial-only subgroup). For patients with baseline MRI data available for both SIJ and spine, lesion presence and distribution were further analysed across subgroups.

Descriptive statistics were used to characterise the full cohort and subgroups based on peripheral involvement status. Continuous variables were reported as mean $\pm$ SD, while categorical variables were presented as n (%) and compared using the χ^2 test or Fisher’s exact test, as appropriate. Given the subjective nature of clinical enthesitis assessment, a sensitivity analysis was performed, excluding enthesitis from the definition of peripheral involvement. Statistical significance was set at a 5% Type I error level. All analyses were performed using SPSS for Windows, V.28.0 (IBM, Armonk, New York, USA).

Table 1 Demographic and clinical characteristics of patients diagnosed with axial spondyloarthritis at baseline by a local rheumatologist, with available MRI data

	n=199 (SIJ MRI data)	n=67 (spinal MRI data)	n=40 (whole spine and SIJ data)
Age, years (mean, SD)	30.3 (9.4)	32.6 (8)	34.5 (7)
Males	109 (54.8)	37 (55.2)	21 (52.5)
Symptom duration, years (mean, SD)	5 (5.8) (n=165)	5.5 (6.2) (n=58)	7.2 (7) (n=34)
HLA-B27 positive	126 (63.3)	46 (68.7)	26 (65)
Elevated CRP	80 (40.2)	26 (38.8)	16 (40)
Psoriasis	18 (9)	5 (7.5)	5 (12.5)
Uveitis	22 (11.1)	13 (19.4)	11 (27.5)
IBD	8 (4)	5 (7.5)	5 (12.5)
Family history	51 (25.6)	18 (26.9)	12 (30)
Arthritis	97 (48.7)	28 (41.8)	17 (42.5)
Dactylitis	12 (6)	5 (7.5)	5 (12.5)
Enthesitis	86 (43.2)	33 (49.3)	15 (37.5)

Data shown are N (%), unless otherwise stated. Column Ns reflect overlapping MRI subsets at baseline. SIJ MRI was available in 199 patients (T2WFS n=199; T1W n=175). Spinal MRI was available in 67 patients; within this, 40 had whole-spine MRI, and 27 had segmental-only MRI. Per-segment counts differ (cervical n=42, thoracic n=40, lumbar n=67). The ‘N=40’ column corresponds to patients with both whole-spine and SIJ MRI at baseline. Counts across columns are overlapping and should not be summed. CRP, C-reactive protein; HLA-B27, Human leukocyte antigen B27; IBD, Inflammatory bowel disease; SIJ, Sacroiliac joint.

RESULTS

Demographic and clinical characteristics

A total of 278 patients were included in the ASAS-CC cohort. Of these, 199 individuals were diagnosed with axSpA at baseline and underwent MRI assessment, forming the study population (table 1, figure 1). SIJ MRI data were available for all 199 patients, including T2WFS sequences in all cases and T1W sequences in 175 cases. Spinal MRI data were available in 67 patients, comprising cervical, thoracic and lumbar scans in 42, 40 and 67 individuals, respectively. Forty patients underwent imaging of the entire spine, while 27 had MRI limited to specific spinal segments, both using T2WFS and T1W sequences.

The mean age of patients was 30 years (SD $\pm$ 9), with 55% being male. Mean symptom duration was 5 years (SD $\pm$ 5.8). HLA-B27 positivity was observed in 63%, while 40% had elevated CRP at baseline. Arthritis, enthesitis and dactylitis were reported in 49%, 43% and 6% of patients, respectively. Psoriasis was present in 9% of cases. Mean age was similar between subgroups with and without spinal MRI assessment, with the only notable difference being a slightly higher prevalence of extra-articular manifestations among those assessed with spinal MRI (table 1). At baseline, none of the patients were

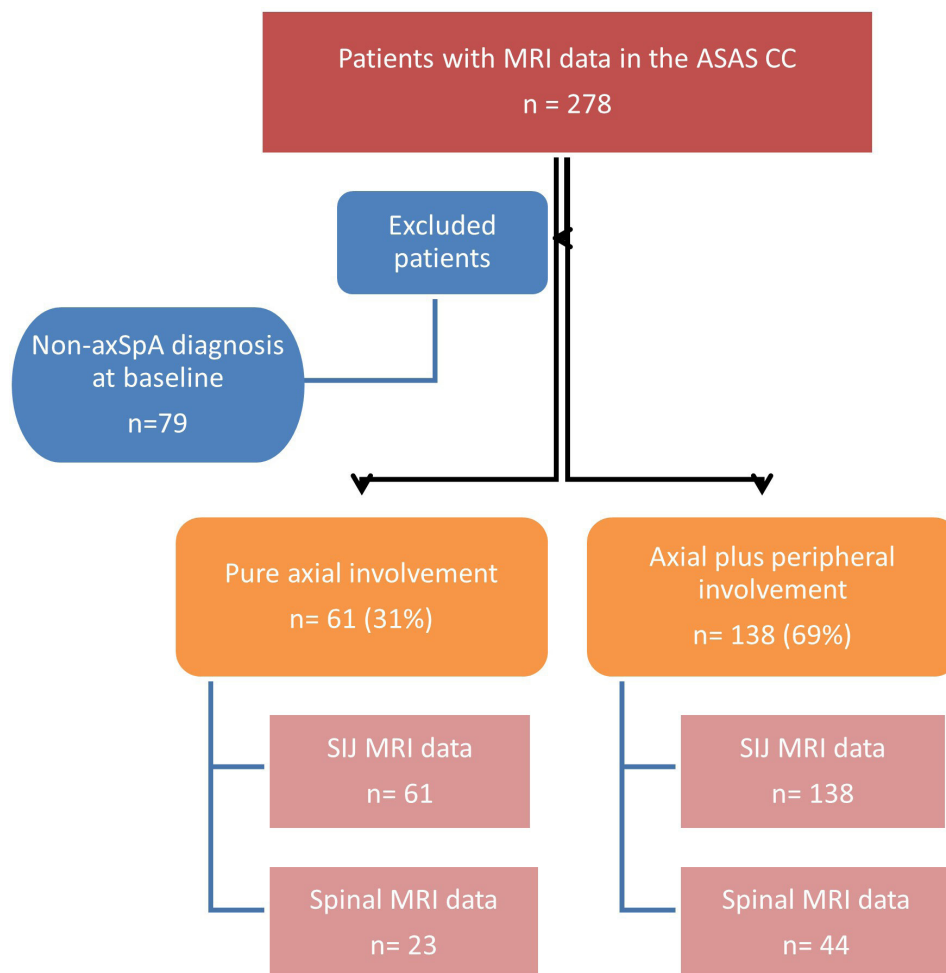


Figure 1 Study flowchart. The numbers shown reflect the availability of MRI sequences at baseline; spinal MRI includes both whole-spine and segmental acquisitions. axSpA, Axial spondyloarthritis; ASAS CC, Assessment of SpondyloArthritis International Society Classification Cohort; SIJ, sacroiliac joints.

being treated with biologic or targeted synthetic disease modifying anti-rheumatic drugs.

Inflammatory lesions of the sacroiliac joints (SIJ)

Data regarding the presence and location of inflammatory lesions of the SIJs are presented in table 2 and figure 2. Subchondral SIJ BMO was detected in 49% of patients, with no quadrant-specific preference. No significant differences were found between axSpA subgroups (54% in the axial-only subgroup vs 47% in the axial-plus-peripheral subgroup), except for sacral involvement, where left sacrum BMO was more prevalent in the axial-only subgroup (34% vs 20%, $p=0.02$).

Capsulitis, inflammation at the site of erosion, and joint space fluid were observed in 4%, 6% and 13% of patients, respectively, with no significant subgroup differences. Enthesitis was recorded in 7% of cases, again with no notable differences between subgroups. Joint space enhancement was evaluated only in the contrast-enhanced subset ($n=48$); none showed enhancement.

Structural lesions of the sacroiliac joints

Data regarding the presence and location of structural lesions of the SIJs are presented in table 2 and figure 2.

Erosion was the most prevalent structural lesion, affecting 35% of patients. Iliac involvement was more than three times as frequent as sacral involvement (27% vs 9%). No significant subgroup differences were observed (43% in the axial-only subgroup vs 32% in the axial-plus-peripheral subgroup), though right iliac erosions were significantly more common in the axial-only subgroup (34% vs 18%, $p=0.02$).

Fat lesion was the second most prevalent structural lesion, identified in 22% of patients. No statistically significant subgroup differences were noted (30% in the axial-only subgroup vs 19% in the axial-plus-peripheral subgroup). However, left sacral fat lesions were more frequent in the axial-only subgroup (23% vs 9%).

Fat metaplasia within erosion cavities was seen in 9% of patients, with no subgroup differences (9% vs 8%). Sclerosis was present in 18%, predominantly in the ilium (14%), with minimal sacral involvement (<1%).

Ankylosis was infrequent, observed in only 3% of patients, with no subgroup distinctions. Non-bridging bone buds were not identified in any cases.

Table 2 Prevalence of inflammatory and structural lesions of the sacroiliac joints in patients with axial spondyloarthritis, and comparison of these lesions between axSpA patients with and without peripheral involvement (defined as the presence of arthritis, dactylitis or enthesitis)

	All patients	Axial only subgroup	Axial+peripheral subgroup	P value
Inflammatory lesions	n=199	n=61	n=138	
Subchondral bone marrow oedema (in any quadrant)	98 (49.2)	33 (54.1)	65 (47.1)	0.36
Right ilium	62 (31.2)	20 (32.8)	42 (30.4)	0.74
Right sacrum	59 (29.6)	21 (34.4)	38 (27.5)	0.33
Left ilium	59 (29.6)	19 (31.1)	40 (29)	0.76
Left sacrum	48 (24.1)	21 (34.4)	27 (19.6)	0.02
Capsulitis (in any side)	8 (4)	3 (4.9)	5 (3.6)	0.7
Right	5 (2.5)	1 (1.6)	4 (2.9)	1.0
Left	4 (2)	3 (4.9)	1 (<1)	0.08
Joint space enhancement (in any quadrant) (n=48)	0	0	0	NA
Right	0	0	0	NA
Left	0	0	0	NA
Inflammation at the site of erosion (in any quadrant)	12 (6)	3 (4.9)	9 (6.5)	0.76
Right ilium	4 (2)	1 (1.6)	3 (2.2)	1.0
Right sacrum	1 (<1)	0	1 (<1)	1.0
Left ilium	8 (4)	1 (1)	7 (5.1)	0.44
Left sacrum	1 (<1)	1 (1.6)	0	0.31
Enthesitis (in any side)	14 (7)	5 (8.2)	9 (6.5)	0.77
Joint space fluid (in any side)	26 (13.1)	13 (12.4)	16 (9.4)	0.44
Right	15 (7.5)	6 (9.8)	9 (6.5)	0.4
Left	20 (10.1)	9 (14.8)	11 (8)	0.14
Structural lesions	n=175	n=53	n=122	
Erosion (in any quadrant)	62 (35.4)	23 (43.4)	39 (32)	0.15
Right ilium	40 (22.9)	18 (34)	22 (18)	0.02
Right sacrum	14 (8)	4 (7.5)	10 (8.2)	1.0
Left ilium	47 (26.9)	18 (34)	29 (23.8)	0.16
Left sacrum	16 (9.1)	7 (13.2)	9 (7.4)	0.26
Fat lesion (in any quadrant)	39 (22.3)	16 (30.2)	23 (18.9)	0.10
Right ilium	15 (8.6)	6 (11.3)	9 (7.4)	0.39
Right sacrum	25 (14.3)	11 (20.8)	14 (11.5)	0.11
Left ilium	19 (10.9)	8 (15.1)	11 (9)	0.24
Left sacrum	23 (13.1)	12 (22.6)	11 (9)	0.01
Fat lesion ≥1 cm (in any quadrant)	10 (10.9)	8 (15.1)	11 (9.1)	0.24
Right ilium	4 (2.3)	3 (5.7)	1 (<1)	0.09
Right sacrum	11 (6.3)	3 (5.7)	8 (6.6)	1.0
Left ilium	4 (2.3)	2 (3.8)	2 (1.7)	0.59
Left sacrum	14 (8)	6 (11.3)	8 (6.6)	0.36
Fat metaplasia in an erosion cavity (backfill) (in any side)	15 (8.6)	5 (9.4)	10 (8.2)	0.78
Right	8 (4.6)	2 (3.8)	6 (4.9)	1.0
Left	10 (5.7)	3 (5.7)	7 (5.7)	1.0
Sclerosis (in any quadrant)	32 (18.3)	9 (17)	23 (18.9)	0.77
Right ilium	24 (13.7)	8 (15.1)	16 (13.1)	0.73

Continued

Table 2 Continued

	All patients	Axial only subgroup	Axial+peripheral subgroup	P value
Right sacrum	1 (<1)	0	1 (<1)	1.0
Left ilium	23 (13.1)	7 (13.2)	16 (13.1)	0.99
Left sacrum	1 (<1)	0	1 (<1)	1.0
Ankylosis (in any side)	6 (3.4)	1 (1.9)	4 (4.1)	0.67
Right	3 (1.7)	1 (1.9)	2 (1.6)	1.0
Left	5 (2.9)	1 (1.9)	4 (3.3)	1.0
Non-bridging bone buds (in any quadrant)	0	0	0	NA
Right ilium	0	0	0	NA
Right sacrum	0	0	0	NA
Left ilium	0	0	0	NA
Left sacrum	0	0	0	NA

Data shown are N (%). Joint-space enhancement reflects only cases with contrast-enhanced sacroiliac joint MRI (n=48). Statistically significant differences highlighted in bold.
NA, Not applicable.

Inflammatory lesions of the spine

Data regarding the presence and location of inflammatory lesions of the spine are presented in [table 3](#) and [figure 3](#). Spinal BMO was detected in 38% of participants with whole spinal MRI data available. Considering individual spinal segments with MRI data available, BMO predominantly affected the lumbar and thoracic vertebrae (31% and 28%, respectively), with no BMO lesions detected at the cervical spine level.

In participants with MRI of whole spine available, BMO was more prevalent in the axial-only subgroup (56%) compared with the axial-plus-peripheral subgroup (25%, $p=0.046$). Considering individual spinal segments with MRI data available, BMO had a prevalence of 48% versus 23% at the lumbar spine level ($p=0.035$) and 44% versus 17% at the thoracic spine level ($p=0.08$), in terms of subgroup comparison (axial-only vs axial-plus-peripheral subgroup).

Structural lesions of the spine

Data regarding structural lesions of the spine are presented in [table 3](#) and [figure 3](#). Fat lesions were the most prevalent spinal structural abnormalities, occurring in 25% of participants with MRI of whole spine available. Considering individual spinal segments with MRI data available, fat lesions primarily affected the lumbar and thoracic spine (18% for both segments), with cervical involvement in 2% of cases. Their prevalence was numerically higher (non-significant p value of 0.059) in the axial-only subgroup (44%) versus the axial-plus-peripheral subgroup (13%). Syndesmophytes or ankylosis were recorded in 5% of patients, whereas no erosions were identified among those with whole spinal MRI data available.

MRI findings in patients undergoing both whole spine and SIJ imaging

Data regarding MRI findings in patients that underwent both whole spine and SIJ imaging are presented in [table 4](#) and [figure 4](#). MRI data encompassing the entire spine and SIJs were available for inflammatory lesion analysis in 40 patients and for structural lesion assessment in 37 patients.

Inflammatory lesions were present in both SIJs and spine in 18% of patients, SIJs alone in 38%, and spine alone in 20%. No significant differences were observed between subgroups ([table 4](#) and [figure 4](#)).

Structural lesions were found in both SIJs and spine in 19% of cases, SIJs only in 30%, and spine only in 5%, again with no subgroup differences ([table 4](#) and [figure 4](#)).

Given the small numbers and the cross-sectional design, we did not undertake additional per-patient cross-tabulations of inflammatory and structural lesions within this subset.

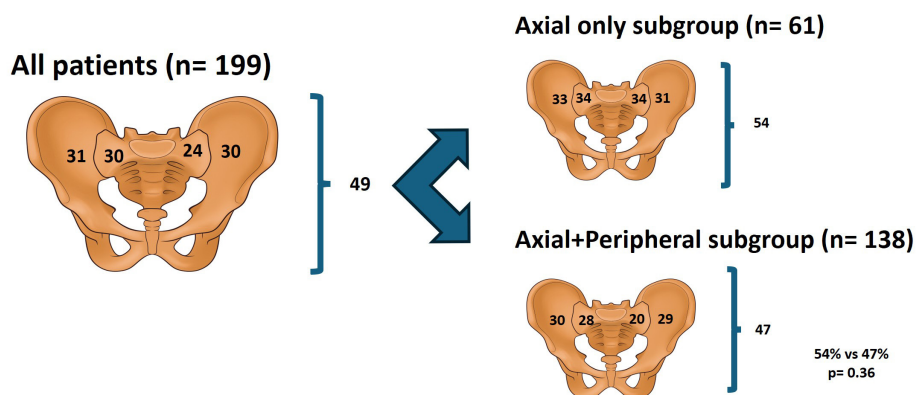
Sensitivity analysis

A sensitivity analysis excluding enthesitis from the definition of peripheral involvement yielded findings comparable to those described above (online supplemental tables 1 and 2).

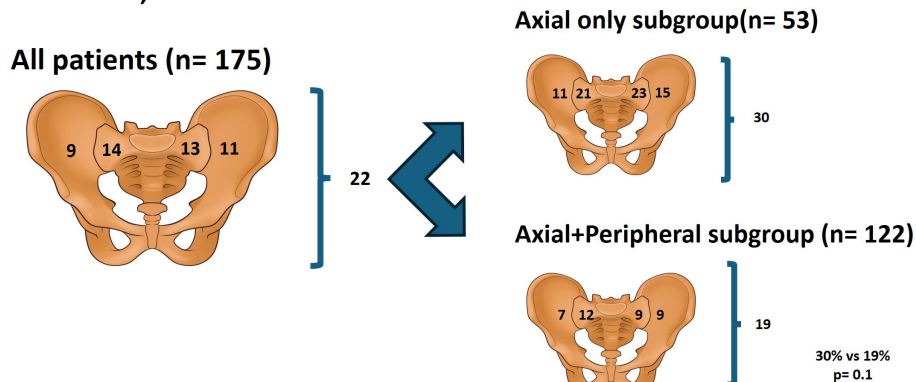
DISCUSSION

This study assessed the prevalence and anatomical distribution of inflammatory and structural MRI lesions in axSpA, and compared patients with isolated axial involvement to those with combined axial and peripheral involvement. BMO was the most common active lesion in the SIJs (49%) while erosions (35%) and fat lesions (22%) were the most frequent structural changes, with erosions and sclerosis occurring more commonly on the

A Subchondral bone marrow inflammation, %



B Fat lesions, %



C Erosions, %

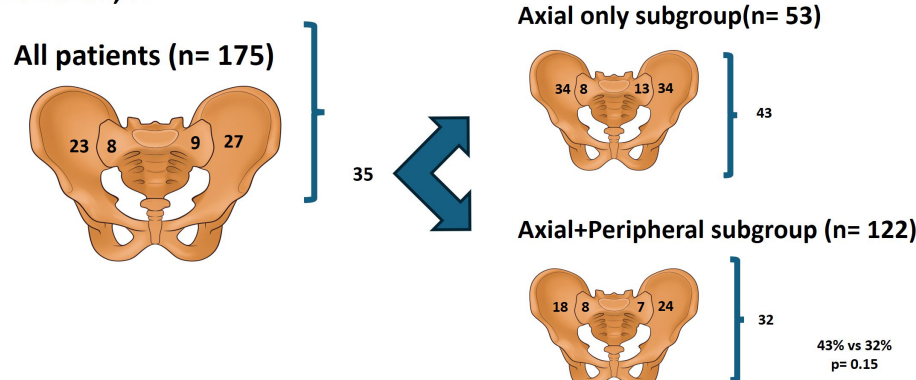


Figure 2 Frequency of common sacroiliac joint lesions identified in the study: subchondral bone marrow inflammation (A), fat lesions (B) and erosions (C).

iliac side. In the spine, BMO, fat lesions and syndesmo-
phytes/ankylosis were present in 38%, 25% and 5% of
participants, respectively. Among those with both SIJ and
whole-spine MRI, inflammatory lesions were identified in
both areas in 18% of patients, in the SIJ only in 38%, and
in the spine only in 20%. This suggests that a subset of
patients may present with isolated spinal inflammation
at diagnosis, which could be missed if imaging is limited
to the SIJs. Structural lesions were found in both areas in
19%, the SIJ alone in 30%, and the spine alone in 5%.
Lesion distribution was similar across axSpA subgroups,
suggesting a broadly comparable MRI phenotype regard-
less of peripheral involvement. The lack of subgroup
differences in BMO distribution suggests that peripheral

involvement may not significantly influence early axial
inflammation patterns and supports the notion of a
shared pathophysiological pathway across axSpA pheno-
types.

Previous studies have seldom examined the prevalence
or anatomical distribution of SIJ and spinal lesions in
newly diagnosed axSpA patients. In this cohort, inflam-
matory and structural abnormalities were observed
throughout the SIJs and spine. SIJ BMO was equally
distributed across the iliac and sacral regions and consis-
tent between subgroups. The prevalence of BMO (49%)
aligns with earlier studies, such as those by Seven *et al*¹⁵
(prospective cohort, n=204; axSpA subgroup mean age
30.9 years, 63% male, TNFi-naïve; SIJ scoring: SPARCC)

Table 3 Prevalence of inflammatory and structural lesions of the spine in patients with axSpA, and comparison of these lesions between axSpA patients with and without peripheral involvement (defined as the presence of arthritis, dactylitis or enthesitis)

	All patients	Axial only subgroup	Axial+peripheral subgroup	P value
Bone marrow oedema				
Cervical segment	0 (n=42)	0 (n=16)	0 (n=26)	NA
Thoracic segment	11 (27.5) (n=40)	7 (43.8) (n=16)	4 (16.7) (n=24)	0.08
Lumbar segment	21 (31.3) (n=67)	11 (47.8) (n=23)	10 (22.7) (n=44)	0.035
BMO in any spinal segment (in participants with MRI of whole spine available)	15 (37.5) (n=40)	9 (56.3) (n=16)	6 (25) (n=24)	0.046
Fat lesions				
Cervical segment	1 (2.4) (n=42)	0 (n=16)	1 (3.8) (n=26)	1.0
Thoracic segment	7 (17.5) (n=40)	5 (31.3) (n=16)	2 (8.3) (n=24)	0.09
Lumbar segment	12 (17.9) (n=67)	6 (26.1) (n=23)	6 (13.6) (n=44)	0.31
Fat lesions in any spinal segment (in participants with MRI of whole spine available)	10 (25) (n=40)	7 (43.8) (n=16)	3 (12.5) (n=24)	0.059
Erosions				
Cervical segment	0 (n=42)	0 (n=16)	0 (n=26)	NA
Thoracic segment	0 (n=40)	0 (n=16)	0 (n=24)	NA
Lumbar segment	1 (1.5) (n=67)	0 (n=23)	1 (2.3) (n=44)	1.0
Erosion in any spinal segment (in participants with MRI of whole spine available)	0 (n=40)	0 (n=16)	0 (n=24)	NA
Syndesmophytes or Ankylosis (SAS)				
Cervical segment	1 (2.4) (n=42)	0 (n=16)	1 (3.8) (n=26)	1.0
Thoracic segment	1 (2.5) (n=40)	0 (n=16)	1 (4.2) (n=24)	1.0
Lumbar segment	3 (4.5) (n=67)	1 (4.3) (n=23)	2 (4.5) (n=44)	1.0
SAS in any spinal segment (in participants with MRI of whole spine available)	2 (5) (n=40)	1 (6.3) (n=16)	1 (4.2) (n=24)	1.0

Data shown are N (%). Statistically significant differences highlighted in bold.
SAS: Syndesmophytes or Ankylosis. axSpA, axial spondyloarthritis; BMO, Bone marrow oedema; NA, Not applicable.

and Hecquet *et al.*¹⁶ (retrospective study, n=200; SpA group mean age 37.4 years, 54% male; SIJ scoring: ASAS MRI Working Group lesion definitions) although subgroup-specific distributions have not been previously reported. Additional types of inflammatory lesions were noted in up to 13% of patients, but no significant differences in location or frequency were observed between subgroups. Given the limited data on inflammatory lesions beyond BMO, further research is needed to clarify their diagnostic and subclassification value.

Among structural SIJ lesions, erosions, fat lesions and sclerosis were the most common, in line with existing reports.^{15 17–19} Erosions (35%) and sclerosis (18%) were more frequently located on the iliac side. No significant subgroup differences were observed. A French cohort reported higher prevalence rates—60% for erosions, 34% for fat lesions and 36% for sclerosis—and noted fat lesions predominantly in sacral quadrants, while erosions and sclerosis were more frequent in the ilium.¹⁶ These discrepancies may reflect variation in patient populations, disease duration, lesion definitions, MRI scoring

methods and how the SIJs were subdivided. This underscores the importance of standardising lesion definitions and approaches in MRI research to enable valid cross-cohort comparisons.

Spinal abnormalities were frequently observed. BMO affected the lumbar and thoracic spine most often (31% and 28%), with lumbar involvement more frequent in axial-only disease (48% vs 25%). Fat lesions (25%) were the most common spinal structural lesion (25%), evenly distributed across thoracic and lumbar levels, while erosions and syndesmophytes were rare. Previous studies^{17 20 21} have demonstrated substantial variability in the detection of spinal lesions on MRI, influenced by study design, disease duration, lesion definitions and other methodological factors. One study²² identified corner inflammatory lesions in 50% of patients with long-standing axSpA, while another²³ reported fat corner lesions in the cervical, thoracic and lumbar spine at rates of 12%, 38% and 36%, respectively. Additionally, a separate study²⁴ investigating spinal MRI in early peripheral

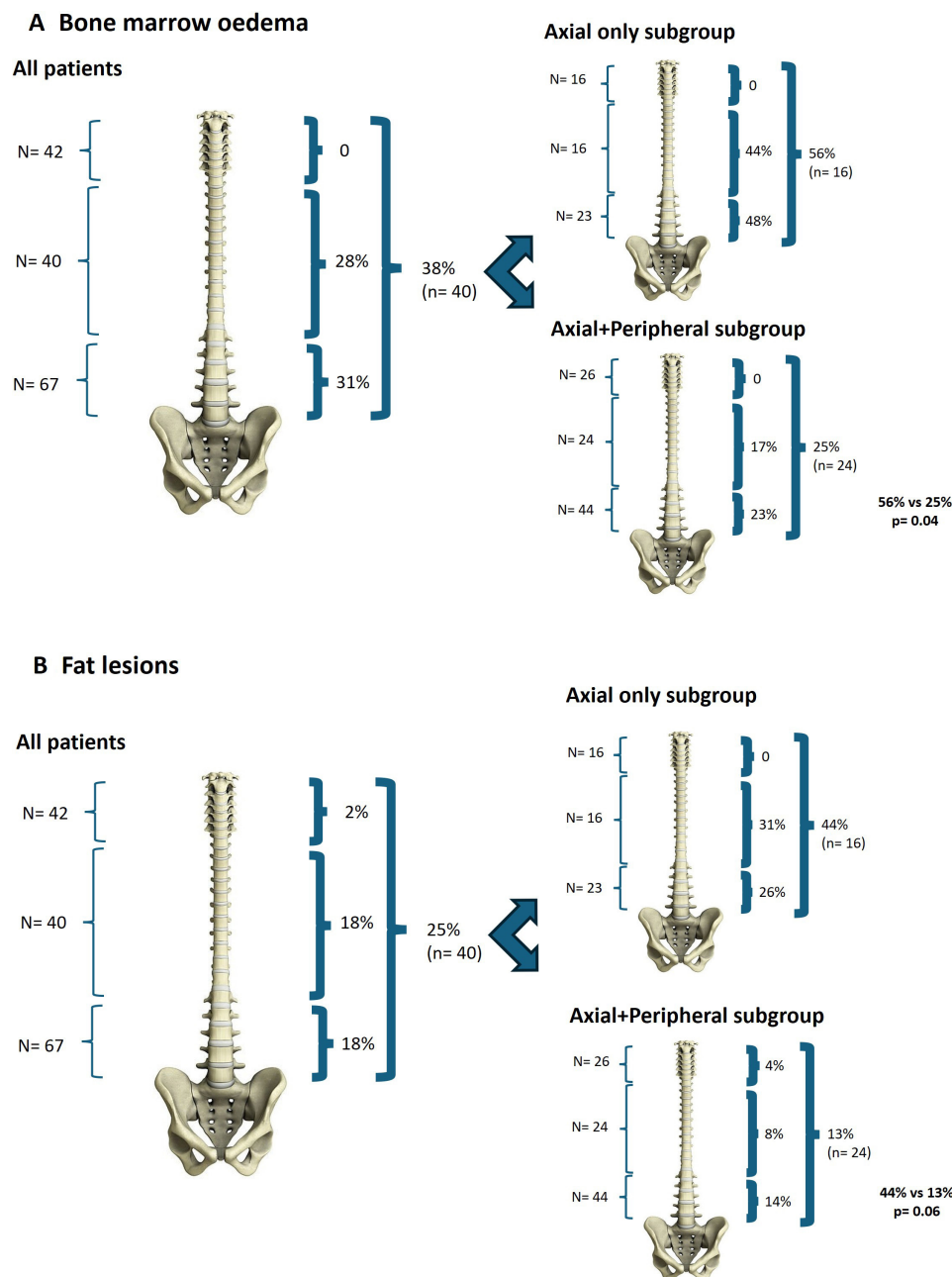


Figure 3 Frequency of common spinal lesions identified in the study: bone marrow oedema (A) and fat lesions (B).

SpA found fat lesions to be the most frequent structural lesion at baseline, observed in 32% of patients.

In clinical practice, spinal MRI is often used to support diagnostic and treatment decisions in axSpA, even though the latest EULAR guidelines do not recommend its routine use for monitoring treatment response or disease activity, due to limited supporting evidence and concerns about cost.¹³ Our findings suggest that spinal MRI may offer additional value, warranting further investigation. Spinal MRI may be particularly useful in selected scenarios where there is a high clinical suspicion of axial SpA but SIJ MRI results are inconclusive. Notably, 20% of patients had inflammatory spinal lesions without SIJ involvement, and 5% had structural lesions confined to the spine, regardless of subgroup. Previous

evidence on this topic remains contradictory. Two studies reported that combined spinal and SIJ MRI did not offer significant advantages over SIJ MRI alone, either because spinal inflammation was rare in the absence of SIJ involvement,²⁵ or because combined MRI resulted in a high rate of false positives.²⁰ In contrast, three studies found spinal inflammation in up to half of patients without SIJ inflammation, suggesting that spinal MRI may provide added value.^{26–28} Given the cross-sectional design, the axSpA-only cohort and the limited number with whole-spine MRI, these observations should be interpreted cautiously and do not imply routine use of spinal MRI. Spinal BMO is not specific and has multiple differential diagnoses. However, findings from this study add to previous evidence suggesting that spinal MRI may

Table 4 Prevalence of inflammatory and structural lesions in the sacroiliac joints and spine in patients with both available sacroiliac joints and spine MRI data at baseline

	All patients	Axial only subgroup	Axial+peripheral subgroup	P value
Inflammatory lesions	n=40	n=16	n=24	
Neither SIJ nor spinal lesions	10 (25)	2 (12.5)	8 (33.3)	0.26
Both SIJ and spinal lesions	7 (17.5)	4 (25)	3 (12.5)	0.41
Only SIJ lesions	15 (37.5)	5 (31.3)	10 (41.7)	0.51
Only spinal lesions	8 (20)	5 (31.5)	3 (12.5)	0.23
Structural lesions	n=37	n=15	n=22	
Neither SIJ nor spinal lesions	17 (45.9)	6 (40)	11 (50)	0.55
Both SIJ and spinal lesions	7 (18.9)	4 (26.7)	3 (13.6)	0.41
Only SIJ lesions	11 (29.7)	3 (20)	8 (36.4)	0.47
Only spinal lesions	2 (5.4)	2 (13.3)	0	0.16

*Peripheral subgroup defined as the presence of arthritis, dactylitis, or enthesitis. Data shown are N (%).
SIJ, sacroiliac joints.

have incremental diagnostic and monitoring value in axSpA.²⁹ It also aligns with growing data indicating that spinal involvement without sacroiliitis is common in axial psoriatic arthritis.^{30 31}

However, the diagnostic utility of spinal MRI is best assessed across varied lesion-count thresholds, and against control groups mimicking axSpA. Such comparative analyses are essential for estimating the sensitivity and specificity of spinal MRI. Imaging must ultimately be interpreted in context, as the diagnosis of axSpA relies on the totality of evidence (signs and symptoms, laboratory results, genetic markers and imaging). Establishing the incremental value of spinal MRI for monitoring requires prospective longitudinal studies with serial imaging and standardised disease activity and treatment measures, to determine whether it adds value beyond SIJ imaging for tracking progression or informing treatment. Future research should explore whether early spinal involvement, particularly in the absence of SIJ lesions, represents a clinically distinct phenotype or predicts differential treatment responses. Such insights could prove valuable in refining classification criteria and optimising imaging strategies. These issues were outside our aims; we draw no firm conclusions on diagnostic or monitoring value.

An international consensus on a standardised image acquisition protocol for diagnostic evaluation of the SIJs by MRI was published by ASAS and SPARTAN in 2024.¹⁴ The main differences compared with previous recommendations include the addition of an SIJ semiaxial T2W sequence with fat suppression and an SIJ semicoronal thin-slice T1W sequence optimised for erosion detection. The findings of this study would likely have been broadly similar if the updated acquisition protocol had been used. The semiaxial T2W sequence with fat suppression could have enabled readers to more confidently distinguish BMO related to axSpA from BMO due to biomechanical or degenerative causes, while the semicoronal thin-slice T1W (erosion-sensitive) sequence could have improved

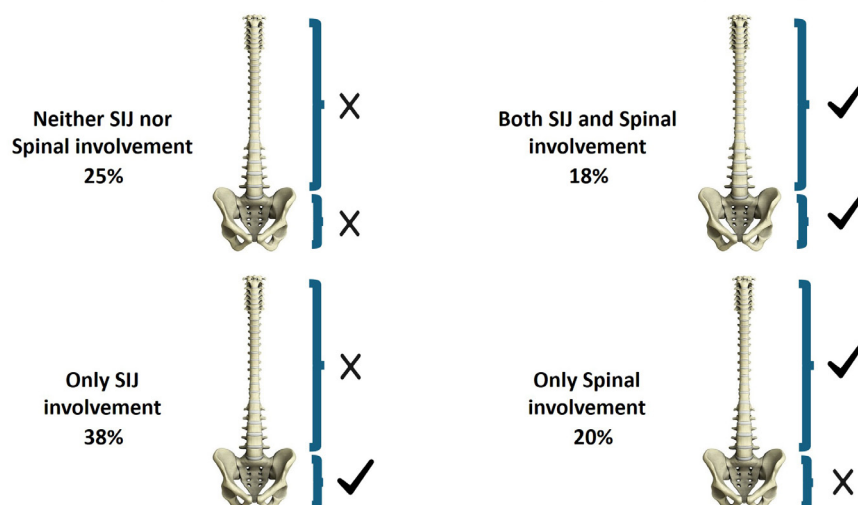
sensitivity for detecting erosions. However, many patients who underwent whole-spine MRI also had high-quality SIJ MRI that included an axial sequence. Consequently, the absence of an updated SIJ MRI protocol is unlikely to have had a significant impact on the interpretation of BMO in this subset. While detection of erosions would likely have been more sensitive with an updated protocol, the overall effect of this is less certain.

Our scans were read centrally by multiple experienced ASAS readers with majority agreement to maximise reliability. In day-to-day practice, local readings can diverge from central assessments. Evidence from the ASAS central-versus-local reader analysis in the ASAS-CC cohort showed that local readers overcalled active SIJ lesions by ~30% compared with central readers; however, this produced only a minor change in ASAS classification status (6.4%), and substituting radiography with MRI structural lesions changed classification in 1.4% of cases.³² This suggests that although absolute lesion frequencies may be inflated under local reading, the overall clinical/classification impact is modest. In light of this, we would expect routine local reading to yield somewhat higher apparent prevalences of active SIJ lesions (and potentially lower specificity) than our centrally derived estimates. Careful application of ASAS lesion definitions, use of example atlases and periodic calibration or second reads for equivocal scans can mitigate this effect.³²

A major strength of this study is the application of ASAS definitions for anatomical mapping of inflammatory and structural MRI lesions, combined with multi-reader consensus scoring. The comparison of axial-only and axial plus peripheral subgroups provides insight into the influence of peripheral involvement on imaging features. Inclusion of patients with both SIJ and whole-spine imaging at baseline allows comprehensive evaluation of lesion distribution across the axial skeleton.

This study has several limitations that should be acknowledged. First, the absence of a control cohort

A Inflammatory lesions – patients with SIJ and whole spine imaging data (n= 40)



B Structural lesions – patients with SIJ and whole spine imaging data (n= 37)

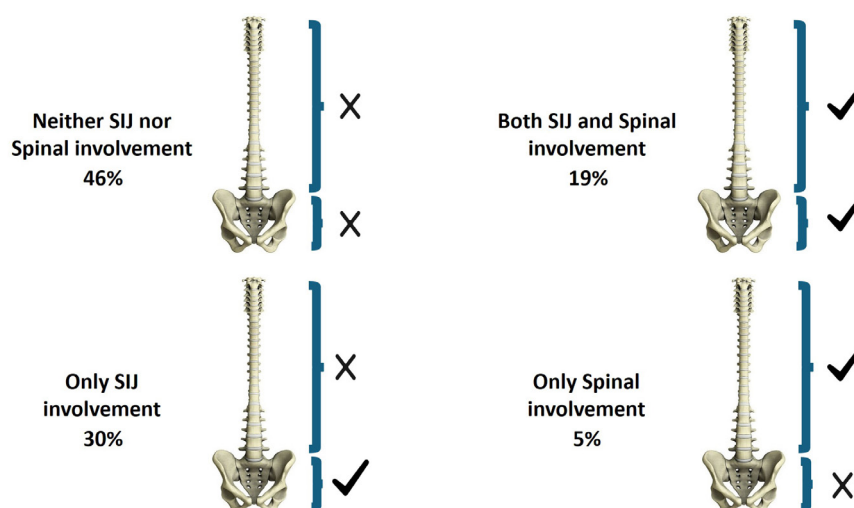


Figure 4 Distribution of sacroiliac and spinal involvement in patients with MRI-confirmed inflammatory (A) and structural (B) sacroiliac and spinal lesions.

limits the ability to assess the diagnostic performance of MRI findings and to distinguish features specific to axSpA from those observed in other conditions. Second, SIJ MRI protocols were not entirely consistent across all patients; for example, some scans included axial sequences while others did not, potentially introducing variability in lesion detection and interpretation. Third, the anatomical extent of spinal MRI varied between patients, which may have influenced the observed prevalence and distribution of spinal lesions. This heterogeneity complicates direct comparisons and may obscure patterns of early spinal involvement. Fourth, the presence of coexistent disc degeneration, a common confounder in spinal imaging, was not systematically recorded, which could affect the interpretation of inflammatory lesions and their attribution to axSpA. Fifth, the study lacks a longitudinal component, limiting the ability to evaluate the monitoring and

prognostic value of spinal MRI findings over time. Sixth, we did not present upper-lower subregional tabulations (eg, within ilium/sacrum or individual spinal levels), which may be informative in specific populations; larger, purpose-designed cohorts are warranted for such analyses. Finally, the number of participants with whole-spine MRI was modest, limiting the precision of subgroup estimates.

In conclusion, this study mapped the distribution of SIJ and spinal MRI lesions at diagnosis in patients with axSpA, using ASAS lesion definitions and comparing those with and without peripheral involvement. Lesions were common throughout the axial skeleton and did not differ meaningfully between subgroups. These data may aid diagnostic reasoning but are currently insufficient to support reliable subclassification. Further prospective studies are needed to evaluate their monitoring, prognostic and clinical relevance.

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